19. More Random Search

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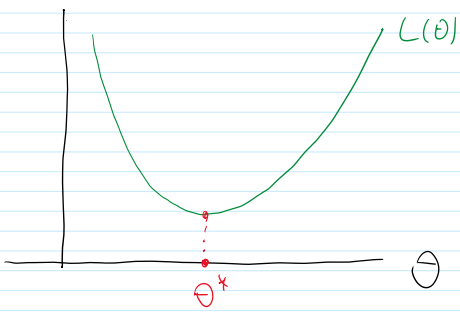
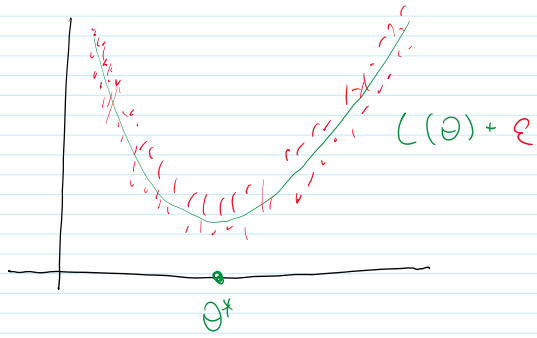
So far, we considered 3 basic search Methods

- SRS : pick points randomly (θ's)
- LRS : pick points near the current best θ
- ELRS : do LRS but include a bias / inertia term to push the search in a potentially good direction.

Sometimes, we don't observe $L(\theta)$
but instead observe $\underbrace{L(\theta)}_{\text{Loss}} + \underbrace{\varepsilon}_{\text{additive Noise}}$





Two Approaches to dealing with Noise

(1) compute the loss Multiple times for each θ

$$\left.\begin{aligned} L_1 &= L(\theta) + \varepsilon_1 \\ L_2 &= L(\theta) + \varepsilon_2 \\ &\vdots \\ L_k &= L(\theta) + \varepsilon_k \end{aligned}\right\} \quad \text{use } \bar{L} = \frac{1}{k}\sum L_i$$

Average k evaluations of the loss function at point θ

- $\mathbb{E}\bar{L} = L(\theta)$
- $Var(\bar{L}) = \dfrac{Var(\varepsilon)}{k}$

Bigger k is the smaller the variance of $\bar{L}$ is.

Pro's : we reduced the variance of the Noise

Con : we now have to evaluate $L(\theta) + \varepsilon$ k times for each θ.

(2) Evaluate once but only update the "best" θ if the noisy loss is less than some threshold

Last time: if $L(\theta_{i+1}) < L(\theta_i)$, we found a better θ so we keep θ_{i+1}

Now, we compare $L(\theta_{i+1}) + \varepsilon_{i+1}$ to $L(\theta_i) + \varepsilon_i$

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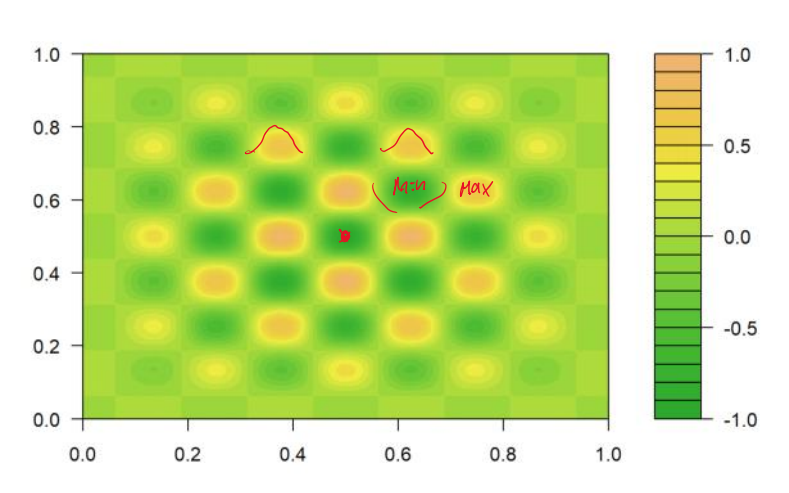
↳ add a "threshold" for identifying an improvement in L

pick some $\tau_i > 0$ and update θ_i to θ_{i+1} if

$$\underbrace{L(\theta_{i+1}) + \varepsilon_{i+1}} < \underbrace{L(\theta_i) + \varepsilon_i} - \tau_i$$

only keep θ_{i+1} if the noisy loss decreases by more than τ.

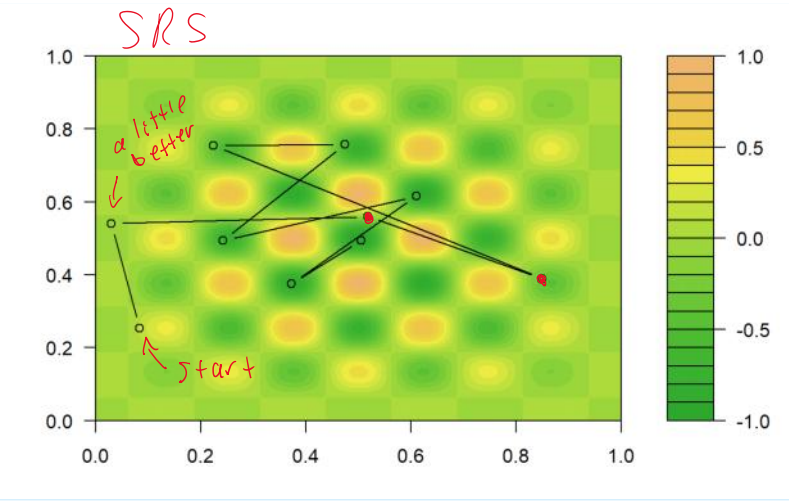


$\theta^* = (\frac{1}{2}, \dots, \frac{1}{2})$

$L(\theta^*) = -1$

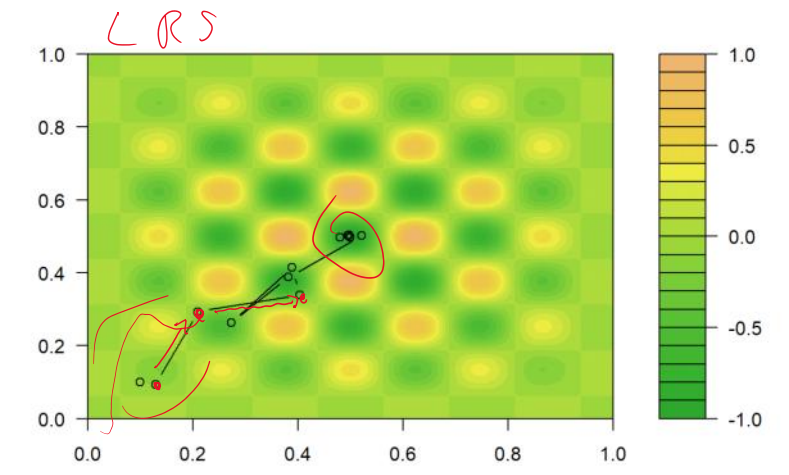
"Curse of Dimensionality"

→ Optimization gets way harder as d gets bigger



← ran 10,000 times

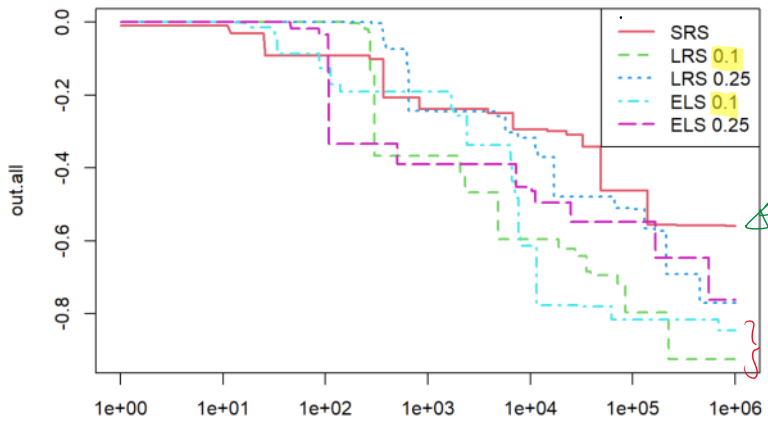
← Purely random choices



← Local choices

Pick a value for the variance of the noise we are adding to θ

6 Dimension,



SRS

LRS
ELS
with $\sigma = 0.1$